

CE 536 - Introduction to numerical methods for civil engineers

Spring 2022 Tue & Thu 3:00 – 4:15 MRC 00136

Description

This is an entry level graduate course intended to give an introduction to widely used numerical methods through application to several civil and environmental engineering problems. The emphasis will be on the breadth of topics and applications; however, to the extent possible, the mathematical theory behind the numerical methods will also be presented. The course is expected to lay foundation for students beginning to engage in their thesis projects that involve numerical methods. Student will use MATLAB as a tool in the course. Experience with MATLAB is not required. The course will be taught in an interactive setting in a computer equipped classroom.

Student Learning Outcomes

Upon completion of the course, the students will be able to:

- Use MATLAB as a programming language for engineering problem solving.
- Describe and apply basic numerical methods for civil engineering problem solving.
- Develop algorithms and programs for solving civil engineering problems involving: (i) multi-dimensional integration, (ii) multivariate differentiation, (iii) ordinary differential equations, (iv) partial differential equations, (v) optimization, and (vi) curve fitting or inverse problems.

Course Organization and Scope

Module	Topic	Duration (75 Min)
1	MATLAB Basics	
1.1	General MATLAB commands and features	Lectures 1-2
1.2	Simple Civil Engineering Examples	Lecture 3
	<i>Assignment 1 – Matlab basics</i>	
2	Numerical Integration and Differentiation	
2.1	Numerical integration techniques and civil engineering applications	Lectures 4-6
	<i>Assignment 2 – Numerical Integration methods with</i>	

	<i>applications</i>	
2.2	Numerical differentiation with applications in groundwater flow	Lecture 7
	<i>Assignment 3 –Advanced numerical integration and differentiation with applications</i>	
3	Ordinary differential equations	
3.1	Runge-Kutta methods with applications in structural and environmental engineering	Lectures 8-9
3.2	Stiff systems with applications in environmental engineering	Lecture 10
3.3	Boundary value problems in structural and environmental engineering	Lectures 11-12
	<i>Assignment 4 – ODE methods and applications</i>	
4	Partial differential equations	
4.1	Introduction to finite-difference methods with civil engineering examples	Lectures 13-15
4.2	Linear and non-linear system solution (direct and iterative methods)	Lecture 16
4.3	Applications in groundwater flow and transport	Lecture 17
4.4	Matlab PDE toolbox	Lecture 18
4.5	Introduction to finite-element methods with application to groundwater flow	Lecture 19
	<i>Assignment 5 – PDE solution of groundwater applications</i>	
5	Optimization	
5.1	Direct methods with applications in environmental and structural engineering	Lecture 20
5.2	Gradient based methods with applications in structural and water resources engineering	Lectures 21-23
5.3	Heuristic methods with applications in water resources engineering	Lectures 24-25
5.4	Design applications in structural and environmental engineering	Lecture 26
	<i>Assignment 6 – Optimization applications in civil engineering</i>	
6	Curve fitting and inverse problems	
6.1	Linear and non-linear regression	Lecture 27
6.2	Direct methods	Lecture 28
6.3	Indirect methods	Lecture 28
6.4	Applications in Civil Engineering	Lecture 29
	<i>Assignment 7- Curve fitting and inverse problems</i>	
	Project assignment due April 23	
	Final Exam (Take Home; Comprehensive) Assigned April 27, Due May 05	

Grading¹:

A weighted average grade will be calculated as follows: Assignments (35%), Quizzes (10%), Project Assignment (20%), Final exam (35%). +/- Grading system will be used.

Policies

Class attendance and full participation is expected of all students and university attendance regulations will be enforced as appropriate. Students shall adhere to the University's policy on academic integrity and classroom conduct found on the university's web site². Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with Disability Services for Students at 1900 Student Health Center, Campus Box 7509, 515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation³.

Prerequisites:

Graduate standing in engineering. Discuss with instructor for any clarification regarding these requirements. Undergraduate students should have a GPA of 3.0 or better and Junior standing.

Recommended Text:

Chapra, S.C., and, R.P. Canale, **Numerical methods for engineers**, McGraw Hill, 5th/6th/7th Edition, 2006/2010/2015.

Optional Texts:

Chapra, S.C. **Applied numerical methods with MATLAB for engineers and scientists**, 3rd edition, McGraw Hill, 2012.

Palm W.J. III, **Introduction to MATLAB 7 for engineers**, 2nd edition, McGraw Hill, 2005.

Schilling, R.J., and S.L. Harris, **Applied numerical methods for engineers using MATLAB and C**, Brooks/Cole Publishers, 2000.

¹ Subject to change

² <http://studentconduct.ncsu.edu/academic-integrity-an-overview>

³ <http://dso.dasa.ncsu.edu/>

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Office Hours:

Thu 4.30 pm – 6 pm via zoom